

Mallikarjun Reddy

SENIOR GENAI ENGINEER | AI SYSTEMS

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PROFESSIONAL SUMMARY

Senior GenAI & AI Systems Engineer with 3+ years architecting autonomous AI products, LLM applications, and high-throughput backend infrastructure. Deep expertise in **RAG architectures**, **Text-to-SQL agents**, **model evaluation**, **parameter-efficient fine-tuning (LoRA/PEFT)**, and **distributed stream processing**. Proven track record of taking AI workflows from design to enterprise GA, scaling platforms to **~20M daily requests**, slashing compute costs by **40%**, and hardening mission-critical cloud services across GKE, PostgreSQL, Kafka, and Celery.

PROFESSIONAL EXPERIENCE

Senior GenAI Engineer | AI Systems | Kredily — Bengaluru, India

Jun 2026 – Present

1. Enterprise GenAI & RAG Intelligence Engine

Python, LLMs, LangChain, RAG, Hybrid Search, PostgreSQL, GKE

- **Key Responsibilities:** Architected conversational GenAI agents and RAG pipelines for HR and payroll intelligence, enabling natural-language querying of complex operational and employee records.
- **Technical Contributions:** Implemented structured schema linking, semantic grounding, multi-turn dialogue management, and prompt safety guardrails to dynamically synthesize ad-hoc analytics without manual dashboarding.
- **Outcome & Impact:** Automated executive summary generation and ad-hoc employee reporting, reducing complex data retrieval turnaround time from hours to instantaneous conversational responses.

2. High-Throughput Platform Performance & DB Scaling

Cloud SQL (PostgreSQL 15), PgBouncer, Redis, GKE, Prometheus

- **Key Responsibilities:** Directed infrastructure scaling and performance tuning for a core enterprise platform serving **~20M requests/day** and **~20k daily active users** with **6k+ peak concurrent sessions**.
- **Technical Contributions:** Diagnosed and remediated database bottlenecks; offloaded heavy analytical read traffic to PgBouncer replicas, implemented proactive Redis nudge-count caching, and resolved ~20 GB month-end disk spill events.
- **Outcome & Impact:** Slashed production peak evening compute demand by **~40% (10 vCPU → 6 vCPU)** and reclaimed **>50% daytime database capacity**.

3. Statutory Payroll Arrears Engine & Attendance Regularisation V2

Django, Celery, PostgreSQL Partitioning, RabbitMQ, Docker

- **Key Responsibilities:** Owned architecture and production rollout of the statutory payroll arrears calculation engine and Attendance Regularisation V2 fleet-wide.
- **Technical Contributions:** Decoupled legacy monolithic models for native PostgreSQL 15 partitioning; automated multi-scenario PF/ESI statutory adjustments, LOP arrears, and bulk EPFO ECR filing generation; revamped Celery background queues with non-blocking exponential backoffs.
- **Outcome & Impact:** Rolled out Attendance V2 across 100% of tenant companies (cutting query volume by ~99%), delivered the arrears engine to GA, and eradicated Celery task timeouts and Redis key eviction spikes.

PROFESSIONAL EXPERIENCE (CONTINUED)

AI and ML Engineer | Isourse Technologies — India Aug 2024 – May 2026

1. Conversational BI & Natural Language Text-to-SQL Platform Python, FastAPI, Hugging Face, vLLM, SQL, LoRA/PEFT, Llama 3, Qwen
 - Key Responsibilities: Built an end-to-end conversational business intelligence platform enabling business teams to explore relational SQL databases using natural-language questions.
 - Technical Contributions: Fine-tuned open-source LLMs (Llama, Qwen, DeepSeek) using SFT and LoRA for Text-to-SQL; implemented AST-based query validation, schema pruning, and self-correction feedback loops.
 - Outcome & Impact: Replaced manual dashboard authoring with interactive querying, achieving >90% SQL execution accuracy on zero-shot domain queries.
2. Enterprise RAG Support Assistant with Persistent Agent Memory LangChain, Milvus, Chroma, FastAPI, Redis, Docker
 - Key Responsibilities: Designed and deployed an enterprise retrieval-augmented customer assistant capable of contextual continuity across multiple user sessions.
 - Technical Contributions: Built semantic chunking, reciprocal rank fusion (RRF) hybrid retrieval (BM25 + dense vector embeddings), and session-state memory persistence in Redis.
 - Outcome & Impact: Decreased customer support ticket resolution turnaround by 35% and drastically minimized hallucination in technical troubleshooting queries.
3. Distributed Stream Processing & Computer Vision Microservices Apache Kafka, PySpark, YOLO, OCR, FaceNet, ONNX Runtime, GKE
 - Key Responsibilities: Built distributed ingestion pipelines and containerized computer vision microservices for identity verification and document extraction.
 - Technical Contributions: Orchestrated real-time streaming pipeline (APIs → Kafka → PySpark → Cloud SQL) and deployed CPU-optimized YOLO + OCR and FaceNet inference pipelines using ONNX Runtime.
 - Outcome & Impact: Processed high-volume distributed telemetry streams and achieved sub-200ms inference latency for automated KYC document parsing.

ML Researcher | Independent / Freelance — Remote Oct 2023 – Aug 2024

- Autonomous Chess-Bot: Built an adversarial game engine combining a custom PyTorch positional evaluator with alpha-beta pruned MiniMax search.
- Air Quality Forecasting Ensemble: Formulated a hybrid time-series forecasting model combining LSTM, GRU, and Facebook Prophet to predict pollutant levels with superior RMSE over individual baseline models.
- Long-Context Language Modeling: Fine-tuned Longformer Encoder-Decoder models (16k context window) for abstractive document summarization using QLoRA and PEFT.

SELECTED PROJECTS & OPEN SOURCE

VisualML Video Generator <i>Gemini API, Remotion, Manim, FFmpeg, Cloud TTS</i> Autonomous engine turning complex topics into narrated animated explainers with synchronized TTS and programmatic video rendering.	[GitHub ↗]	Document AI Chatbot <i>RAG, Local LLMs, Vector Retrieval, Privacy-First</i> On-device question-answering system over private enterprise documents with hybrid retrieval and zero external data transmission.	[GitHub ↗]
Text-to-SQL Model Fine-Tuning <i>PEFT, LoRA, Hugging Face Transformers, Spider Benchmark</i> Fine-tuning harness optimizing lightweight open models to translate multi-table SQL queries with evaluation on standard benchmarks.	[GitHub ↗]	Image Colorization <i>Conditional GANs, PyTorch, Computer Vision</i> Deep generative framework training conditional GANs to reconstruct photorealistic color distributions from grayscale imagery.	[GitHub ↗]

TECHNICAL SKILLS

AI & Machine Learning:	LLMs (Llama/Qwen/DeepSeek), RAG, Agentic Workflows, Prompt Engineering, Model Evaluation, LoRA/PEFT, Text-to-SQL, Computer Vision (YOLO/OCR), Transformers
Systems & Cloud:	Python, C++, SQL, PostgreSQL (Partitioning/Optimization), PgBouncer, Kafka, PySpark, Celery, Redis, RabbitMQ, Docker, Kubernetes (GKE), Cloud SQL, CI/CD

EDUCATION

B.Tech, Mechanical Engineering — Indian Institute of Technology Roorkee 2019 – 2023 | CGPA: 7.4/10 | JEE Adv Rank: 3015